Materials Today Communications 53 (2026) 115294

Contents lists available at ScienceDirect

Materials Today Communications

journal homepage: www.elsevier.com/locate/mtcomm

Multidimensional thermo-mechanical simulation of laser welding dynamics in magnesium–steel joints via hybrid finite element–machine learning frameworks

Gajanan Shankarrao Patange [a], Ganesh Dongre [b], Yogesh Dinkar Jadhav [c], Mohammed Hameeduddin Haqqani [d], Ismoilov Bobur [e], Ankit Tyagi [f,i,*], Muyassar Allaberganova [g], Adilbek Dauletov [h]

[a] Department of Mechanical Engineering, CSPIT–Charusat, Charotar University of Science and Technology, Charusat Campus, Changa, Anand, Gujarat, India
[b] Department of Mechanical Engineering, Vishwakarma Institute of Technology, Pune, Maharashtra 411037, India
[c] Department of Mechanical Engineering, Sinhgad College of Engineering, Savitribai Phule Pune University Pune, Maharashtra, India
[d] Department of Mechanical Engineering, Career Point University, Kota, Rajasthan 325003, India
[e] Department of Preschool and Primary Education, Termez University of Economics and Service, Termez, Uzbekistan
[f] Department of Mechanical Engineering, Poornima College of Engineering, Jaipur, India
[g] Department of Data Transmission Networks and Systems, Urgench State University, Urgench, Uzbekistan
[h] Alfraganus University, Yukori Karakamish Street 2a, Tashkent 100190, Uzbekistan
[i] Center for Innovation and Inclusive Research, Sharda University, Greater Noida, India

ARTICLE INFO

Keywords:
Laser welding
Magnesium–steel joints
Thermo-mechanical simulation
Finite element analysis
Artificial neural network (ANN)
Deep neural network (DNN)

ABSTRACT

Lightweight structural welding of dissimilar magnesium-steel joints Laser welding is essential but faces challenging complex thermo-mechanical behaviour because of material mismatch. This paper introduces a welding dynamics optimization and modeling hybrid finite element (FE) and machine learning system. To model transient temperature field, melt pool development and residual stress development under different laser parameters, a multidimensional thermo-mechanical FE model was developed. Artificial neural networks (ANN) were trained on high-fidelity FE data to make scalar predictory forecasts and deep neural networks (DNN) to make high-dimensional spatiotemporal field predictions. ANN was able to learn nonlinear correlations between process parameters and weld quality measures, whereas DNN was able to reproduce temperature and stress distributions. The thermocouples and infrared thermography Assessed with thermal experiments showed a high degree of agreement with the predictors of FE with a difference of less than 6%. The hybrid framework obtained a coefficient of determination, which was greater than 0.94 and prediction errors of less than 5%. Furthermore, the calculation time was cut about 80% over so-called full FE simulations. It was observed as a result of sensitivity analysis that laser power and scanning speed have a significant impact on thermal gradients, melt pool geometry, and residual stress distribution. The suggested method allows quick prediction, optimization of the process and better quality of the weld. This framework offers a scalable and computationally efficient platform of intelligent laser welding of dissimilar magnesium steel joints and offers future application with real-time monitoring and digital twin systems.

1. Introduction

Laser welding had become an important joining technology of lightweight magnesium-steel structure since it provided high energy density processing ability, excellent control of heat input, and also by the negligible amount of heat-affected zone as compared to the traditional fusion-based welding methods. According to the previous studies, magnesium-steel hybrid materials were being used more in the

* Corresponding author.
E-mail addresses: gajananpatange.me@charusat.ac.in (G.S. Patange), ganesh.dongre@vit.edu (G. Dongre), sittpo3@gmail.com (Y.D. Jadhav), hameed6162@yahoo.com (M.H. Haqqani), bobur_ismoilov@tues.uz (I. Bobur), tyagiankit10506@yahoo.com, ankit.tyagi@poornima.org (A. Tyagi), amuyassar83@gmail.com (M. Allaberganova), a.dauletov@afu.uz (A. Dauletov).

https://doi.org/10.1016/j.mtcomm.2026.115294
Received 26 January 2026; Received in revised form 23 April 2026; Accepted 28 April 2026
Available online 29 April 2026

automotive, aerospace, and transportation systems to ensure weight savings but without compromising the mechanical strength and functional capabilities of the system [1]. Laser welding was also localized and relatively easy to control thus enabling manufacturers to make dissimilar materials which previously could not be welded together through arc-based or resistance welding techniques [2]. Previous studies had shown that laser welding enhanced the efficiency of the joint, minimized distortion, and facilitated the high throughput manufacturing process and thus can be applied in lightweight design at the industry level [3]. Nevertheless, the dissimilar magnesium-steel welding raised significant thermo-mechanical issues because of the wide dissimilarity of physical, thermal and mechanical properties between the two materials. Magnesium had low melting temperature, thermal conductivity and vapour pressure with steel having much higher thermal diffusivity, elastic modulus and melting point [4]. These discontinuities led the heat transfer to be highly nonlinear, and the temperature gradients were steep and the melt pools were formed asymmetrically in the process of laser irradiation [5]. The researchers noted that these conditions tended to lead to an unstable keyhole behavior, intermetallic compounds that were formed in large amounts at the interface, residual stress concentration and cracking caused by solidification [6]. These effects were critical to the integrity of the welds, strength of the joints and the reliability in the long run.

The differing joint that was investigated in this paper was that of AZ31B magnesium alloy and AISI 1018 low-carbon steel. Both materials have temperature-dependent thermal and mechanical properties included in the finite element model, which was used to reproduce nonlinear thermo-mechanical behavioral characteristics in laser welding. The base materials used in this study were AZ31B magnesium alloy and AISI 1018 low-carbon steel because these were the most commonly utilized and thermo-mechanical property data are well documented. AZ31B magnesium alloy was the material of study due to its low density, moderate strength, and good weldability and low melting temperature and high thermal conductivity of the material greatly affected the interaction of the laser and the material. AISI 1018 steel was the dissimilar counterpart since it was able to offer steady mechanical performance, high melting temperature, as well as wide application in structural aspects. The strong difference in thermal and mechanical attributes between the AZ31B and AISI 1018 provided a difficult dissimilar welding environment that facilitated the research to investigate complex issues of thermo-mechanical coupling. The thermal and mechanical characteristics of both materials were taken into account in the finite element model and were temperature-dependent to simulate transient heat transfer, melt pool growth, and residual stress formation during the entire process of welding with lasers in a real-life scenario.

Experimental research has been used to study the morphology of welds, the microstructure, and mechanical performance of magnesium-steel joints. Although experiments were needed to provide a critical validation information, they had to be subjected to a lot of trial and error testing, consumed a lot of material and highly instrumented [7]. Additionally, the measurement of transient temperature fields, melt pool development and stress-strain behavior during laser welding were extremely difficult because the heating rates were very fast and thermal conditions were extremely hostile [8]. Consequently, the multidimensional thermo-mechanical nature of dissimilar laser welding processes could not be completely characterized using experimental methods only. The use of finite element (FE) modeling was hence popular to model laser-material interaction, transient heat transfer, and thermo-elasto-plastic deformation of welding. In the past, FE-based investigations have been able to forecast temperature fields, geometry of the weld pool, and the development of residual stress in different welding geometries [9]. However, precise FE modeling of magnesium–steel laser welding demanded a very fine spatial and temporal discretization, complicated material models that depended on temperature and a detailed description of the phenomena of phase change [10]. The computational cost was greatly increased due to these requirements and only parametric studies, real-time process control, and high speed optimization of the welding parameters were affordable by use of FE models.

To address such limitations, more and more researchers moved towards hybrid finite element-machine learning models. The machine learning methods were proved to be highly able to learn nonlinear association between welding parameters and output quality measurements and provide quick prediction after training [11]. But purely data-based methods tended to be of little physical insight and relied on large experimental datasets, which were challenging to access in complicated welding tasks. Combining FE simulations with machine learning was another viable option, based on physics-based simulations to obtain high quality training data and machine learning models to predict highly complex thermo-mechanical responses with high efficiency. Using artificial neural networks, thermo-mechanical regression tasks were traditionally solved, and deep neural networks were found useful in high-dimensional spatiotemporal features learning when using fields produced by FE models [12]. This mixed modeling model inspired the current work, since it provided a balanced solution that balances physical fidelity, predictive power and computational efficiency of multidimensional modeling of the dynamic process of laser welding in magnesium-steel joints.

Physics-Informed Neural Networks (PINNs) have become one of the promising methods to address thermo-mechanical problems by incorporating the governing equations into the training process of the neural network in recent years. Although PINNs can be very theoretically consistent, their use in complex manufacture tasks like dissimilar metal laser welding can be costly, with sharp thermal gradients, and only scaleable to high dimensional spatiotemporal domains. The proposed hybrid FE-ANN/DNN framework, in its turn, implements the data-driven surrogate modeling approach, resting on the high-fidelity finite element simulations. Rather than addressing governing equations in the training process, the model trains nonlinear mappings on physically consistent simulations data to learn faster and converge to a solution with much less computational effort. This method is especially beneficial to issues with complicated boundary conditions, phase transitions and significant material heterogeneity, as in magnesium steel laser welding.

Whereas PINNs typically involve iteration of partial differential equations as part of the training process, the proposed algorithm separates physics simulation and machine learning, therefore providing efficient training and real-time inference. ANN-based scalar prediction and DNN-based high-dimensional field reconstruction offers a platform of high flexibility, scalability that can deal with both low- and high-dimensional outputs. Hence, the originality of this study is in the combination of finite element physics with multi-level machine learning models to balance physical fidelity, computational efficiency, and scalability to provide a viable alternative to PINN-based models to industrial welding problems.

The main contributions of this research paper are summarized as follows:

- ✓ A multidimensional thermo-mechanical finite element model was developed to accurately simulate transient temperature fields, melt pool dynamics, and residual stress evolution in dissimilar magnesium–steel laser welding.
- ✓ A hybrid physics-informed framework was proposed by integrating finite element simulations with artificial neural network (ANN) and deep neural network (DNN) models to capture nonlinear process–structure relationships efficiently.
- ✓ ANN-based regression and DNN-based high-dimensional learning enabled accurate prediction of weld quality metrics and spatiotemporal thermo-mechanical fields with significantly reduced computational cost.
- ✓ The proposed framework facilitated sensitivity analysis and process optimization of laser parameters, demonstrating its potential for

rapid decision-making and scalable application in intelligent welding systems.

2. Related work

The thermal-mechanical finite element (FE) modeling had been widely applied in the research of the laser welding because it was able to represent the dynamic heat transfer, phase transformation, and stress strain development in rapid thermal cycling. First studies created three dimensional FE models with moving heat source equations to model laser-material interaction and estimate temperature field and weld pool geometry [13]. To make numerical accuracy better, material properties and latent heat including temperature effects were incorporated. Later research generalized these equations to obtain thermo-elasto-plastic behaviour, which facilitated the determination of remanent stresses and distortion caused during welding processes by heating and cooling. Notwithstanding this excellent physical understanding, these FE models necessitated a fine spatial and temporal discretization, which greatly augmented the cost of calculation and constrained their applicability to big scale parametric investigations and in real-time settings [14]. The numerical experimental results of the dissimilar magnesium-steel joint formation showed some other complexities than those of similar materials welding set up. The model in which researchers were able to model the asymmetric heat flow was due to the significant difference in thermal conductivity, melting temperature, and mechanical properties between magnesium and steel. Some experiments have found a problem with the properly modeling melt pool instability, interfacial mixing, and uneven solidification behavior with traditional models of heat sources [15]. Efforts were also made to include intermetallic compound formation and interface reactions but because of finite availability of reliable thermophysical data simplification of assumptions was often needed. These limitations decreased the predictive power, especially in predicting the interfacial stress concentration, crack opening, and defect creation of Mg-steel joints [16].

Modeling of welding processes had increasingly been done using machine learning methods with the improvement of computational power and data availability. Using the parameters of the process, including laser power and welding speed, researchers used the method of artificial neural networks to predict weld bead geometry, penetration depth, and mechanical properties [17]. Support vector machine and random forest algorithms were used to classify the quality of the weld and detect defects in other works. These machine learning models were characterized by high nonlinear learning and fast inference after training [18]. They however were highly dependent on experimentally generated datasets they were not physically interpretable, and their extrapolation was restricted to trained operating conditions [19]. A hybrid approach to physics and data-driven models was seen as a successful approach to the shortcomings of purely numerical and purely data-driven models. Finite element or analytical models produced high-fidelity synthetic datasets in such frameworks which they used to train machine learning algorithms. The scientists proved that physics-informed datasets enhanced the efficiency of learning, decreased the experimental dependency, and maintained physical consistency [20]. The use of hybrid methods in additive manufacturing, machining and arc welding was achieved successfully to forecast thermal history and residual stress with significantly lesser computational effort. Deep neural networks were more and more used to directly learn high-dimensional spatiotemporal field data, like temperature and stress distributions, directly by simulated results, with better accuracy and performance [21].

Regardless of these developments, significant research gaps still existed in the framework of dissimilar laser welding of magnesium-steel joints. The majority of the available hybrid systems concentrated on either similar-material systems or low-dimensional output variables, including scalar measures of the quality of welds. Little research was conducted on the study of high-dimensional thermo-mechanical fields that were unique to Mg–steel welding [22]. Besides that, the joint application of the artificial neural networks to regression and deep neural networks to field level learning in a single framework was not sufficiently studied. Earlier studies have also hardly determined computational savings in comparison with full-scale FE simulations, or provided systematic sensitivity analysis to find process optimization [23]. The current paper has addressed these gaps by coming up with a multidimensional hybrid FEANN/DNN framework in modeling the laser welding dynamics in magnesium-steel joints with the aim of modelling the accurate and computationally efficient dynamics of a scalable and intelligent process optimization [24], [25]. Table 1 presents the previous studies on thermo-mechanical modeling and machine learning-based laser welding. The main models used in existing works to predict the temperature fields, weld pool shape and residual stresses were finite element models, whereas recent studies used machine learning to predict the quality of weld.

3. Multidimensional thermo-mechanical finite element modeling

3.1. Heat transfer and thermo-elasto-plastic equations

A coupled thermo-mechanical finite element model was developed in the study to model the dynamics of laser welding between dissimilar joints of AZ31B magnesium and AISI 1018 steel. The model was then used to solve the transient heat conduction equation to model temperature evolution caused by laser taking into consideration temperature-dependent thermal conductivity, density, and specific heat of the two materials. Latent heat effects were included in the formulation to have melting and solidification in the formation of the weld pool. To model mechanical response, the paper used thermo-elasto-plastic constitutive equations which linked the strains of thermo with elastic and plastic deformation. The model considered the thermal expansion incompatibility between magnesium and steel that resulted in discontinuous stress-strain fields when rapidly heated and cooled.

Temperature-dependent thermal properties and conditions of continuity at the interface between the Mg and steel interfaces were strictly established in the finite element model of asymmetric heat flow across the interface. AZ31B magnesium and AISI 1018 steel have a large disparity in their thermal conductivity and diffusivity, which results in non-uniform heat propagation, resulting in uneven temperature distribution and melt pool development. This was modelled by enforcing numerically solving the time-dependent heat conduction equation in each material domain and continuity of temperature and heat flux forces on the interface, i.e.

$$k_{Mg}\nabla T_{Mg}\cdot n = k_{Steel}\nabla T_{Steel}\cdot n$$

Enabling physically uniform energy transfer between the dissimilar interface.

Moreover, the traveling laser heat source is not acting uniformly on all materials as different materials have different absorptivity and heat reactivity and thus they do not heat uniformly and have irregular geometry of the melt pools. This asymmetry was manifested in the FE results, in skewed temperature contours and uneven depth of melt pool that are well aligned with experimentally observed behavior in dissimilar metal welding. This was made possible through the inclusion of temperature-dependent behavior and interface and interfaces and allowed the representation of phase transformation and local thermal gradients, both of which are important in predicting the residual stress distribution and interfacial failure mechanisms in magnesium steel joints.

The simulation allowed spreading thermal and mechanical governing equations in a single framework that could accurately reflect the formation of residual stresses, distortion, and interfaces stress concentration, which are typical of the dissimilar magnesium-steel laser welding.

Table 1
Summary of related work on thermo-mechanical modeling and ML-based laser welding.

Welding Process / Material System	Modeling Approach	Machine Learning Used	Key Outputs Analyzed	Major Limitations	Relevance to Present Work
Laser welding (similar metals)	3D thermo-mechanical FE	Not used	Temperature, weld pool	High computation cost	Baseline FE methodology
Laser welding (steel–steel)	FE with moving heat source	Not used	Residual stress, distortion	Limited parametric scope	Stress modeling reference
Dissimilar Mg–Al welding	FE thermal model	Not used	Melt pool geometry	Simplified interface modeling	Dissimilar heat flow insight
Mg–steel laser joining	Numerical thermal analysis	Not used	Temperature gradients	No stress prediction	Highlights Mg–steel complexity
Laser welding (various alloys)	Experimental + ANN	ANN	Penetration depth	No physical interpretability	ANN regression benchmark
Laser welding quality prediction	Data-driven modeling	SVM, RF	Weld quality class	Large data requirement	ML-only limitation
Arc welding processes	Hybrid FE–ML	ANN	Thermal history	Low-dimensional outputs	Early hybrid framework
Additive manufacturing	Physics-informed ML	DNN	Temperature fields	Material-specific models	High-dimensional learning
Welding distortion prediction	FE + surrogate ML	ANN	Distortion	Limited field prediction	Surrogate modeling
Laser welding optimization	FE-based parametric study	Not used	Process windows	Time-intensive	Motivation for ML integration

3.2. Laser heat source modeling and boundary conditions

The experiment symbolized the input of laser energy by a moving volumetric model of heat sources to simulate realistically the laser-material interaction in welding. The heat source scanned in the same direction of the welding at a specified scanning rate and provided spatially dispersed energy to both domains of magnesium in addition to steel. In this method, asymmetric heat absorption and melt pool geometry were obtained due to the relative differences in thermal properties of AZ31B and AISI 1018. The interaction with the surrounding environment was modeled by convective and radiational heat losses on the exposed surfaces and adequate thermal contact conditions were imposed on magnesium-steel interface. Mechanical boundary conditions only limited rigid body movement and not the deformation caused thermally. Boundary conditions embraced by the finite element model were simplified to trade off between computational efficiency and physical realism. In real-life welding using lasers, other parameters include clamp forces, fixation limits as well as flow of shielding gases and affect heat transfer and deformation characteristics. Although these effects were not explicitly modeled, their effect was estimated by the constrained displacement boundary conditions and convective heat transfer coefficients used at the exposed surfaces.

The convection coefficients were chosen according to the standard conditions of the welding environment to consider loss of heat to the shielding gas and the ambient conditions. Likewise, mechanical constraints were included to inhibit the rigid body motion and permit the thermally induced deformation, which is approximated to realistic conditions of fixtures. These simplifications have been justified by the fact that there is a tight correlation between FE predictions and experimental data of the fusion zone geometry and thermal behaviour (Table 6), which means that the taken boundary conditions reflect the prevailing thermo-mechanical processes of the welding process. Thus, simplified boundary conditions help in giving a realistic and computationally effective modeling of realistic welding conditions. The respective formulations of the boundaries allowed predicting transient temperature fields, melt pool dynamics and the distribution of residual stresses with high accuracy, which ensure the high level of relevance to the observed thermo-mechanical behaviour of dissimilar laser welding of magnesiumsteel joints.

3.3. Magnesium and steel in different phases and at different temperatures

Detailed material properties of the AZ31B magnesium alloy in the study were temperature-dependent and of AISI 1018 steel to ensure that the study information precisely reflected the unique thermo-mechanical response of each material in terms of laser welding. In the case of AZ31B magnesium, the model used thermal conductivity and specific heat as functions of temperature, the density of the material, elastic modulus, yield strength and thermal expansion coefficient. The model took into consideration large amounts of softening of magnesium at high temperatures which had great effects on the growth of melt pools, concentrated plastic deformations, and stress release in welding. The phase-dependent behavior was modeled by the inclusion of latent heat in the process of melting and solidification and realistic representation of the change between solid and liquid phases in the fusion zone was possible. The Table 2 shows the nominal chemical composition of the AZ31B magnesium alloy and AISI 1018 steel, in which the metals appear enriched with aluminum and iron, respectively, which cause high thermal and metallurgical mismatch in laser welding.

In the case of AISI 1018 steel, the model used temperature dependent mechanical and thermal properties to capture the higher melting temperature, higher stiffness of the alloy and high value of plastic deformation. The steel field had relatively slower thermal diffusion and was stronger throughout a broader temperature field, resulting in only the asymmetrical stress development throughout the dissimilar joint. The model accounted for the phase transformation effects in steel by modifying material properties over temperature ranges of solid state that are pertinent in welding. The strong thermal mismatch, localization of interfacial stress, and residual stress distribution in magnesium–steel laser welding could be modeled explicitly by the model as a phase-dependent behavior in both materials, which increased the predictive quality and physical realism of the model.

Fig. 1 shows heat input induced by laser in the interface of the AZ31B and AISI 1018, and asymmetric thermal conduction, and melt pool formation. It emphasizes the formation of intermetallic compounds and thermal mismatch which determine the formation of residual stress and integrity of the joints in the process of laser welding.

3.4. Mesh strategy and temporal discretization

The experiment also used a three-dimensional non-uniform finite element mesh to provide a resolution of steep thermal gradient and localized mechanical deformation development during laser welding of AZ31B magnesium-AISI 1018 steel joints. A very fine mesh was created in the fusion zone, heat-affected zone and in the material interface in order to measure rapid temperature increase, development of melt pools, and the concentration of interfacial stress. In areas that were far away with the line of the weld, coarser elements were used to minimize computational cost without compromising numerical stability. The transition between fine and coarse areas using the mesh was graded smoothly to prevent the numerical artifacts and convergence. Different

Table 2
Chemical composition of materials used (wt%).

Material	C	Mn	Si	Al	Ni	Cr	Mo	V	Ti	Nb	Fe
AZ31B Magnesium Alloy	0.00	0.30	0.05	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.005
AISI 1018 Steel	0.18	0.75	0.25	0.03	0.02	0.03	0.00	0.00	0.00	0.00	Balance

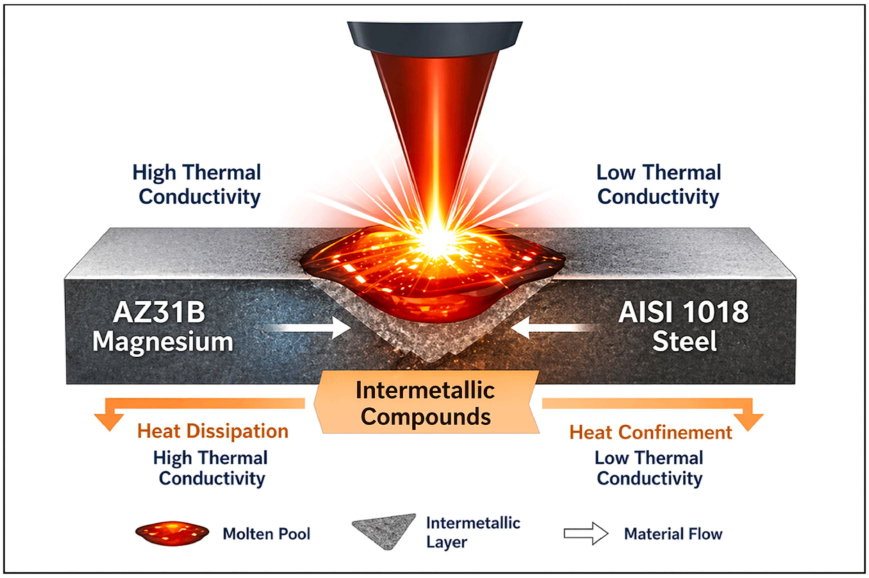

Fig. 1. Thermo-mechanical representation of laser welding in dissimilar magnesium–steel joints.

mesh densities of magnesium and steel domains were used to simulate a different thermal diffusion behaviour and stiffness behaviour.

The computational domain Ω representing the AZ31B–AISI 1018 joint was divided into a finite number of elements Ωe such that:

$$\Omega \quad = \quad \bigcup (e = 1 \quad to\ Ne)\Omega e$$

The temperature and displacement fields were approximated using shape functions:

$$T(x, \quad t) \approx \quad \sum_{i=1}^{n} Ni(x)Ti(t)$$

$$u(x, \quad t) \approx \sum_{i=1}^{n} Ni(x)ui(t)$$

where:

Ni(x) = shape functions
Ti = nodal temperatures
ui = nodal displacements
n = number of nodes per element

Mesh refinement was applied near the weld zone:

$$h\_weld \quad << \quad h\_base$$

where h denotes characteristic element size.

After spatial discretization, the transient heat equation became:

$$CT\left(\frac{dT}{dt}\right) + \quad KT\ T \quad = \quad Q(t)$$

where:

$CT = \int \Omega\, \rho cp\ NT\ N\ d\Omega$ (capacity matrix)
$KT = \int \Omega k(\nabla N)T(\nabla N)d\Omega$ (conductivity matrix)
$Q(t)$ = laser heat input vector

A backward Euler time integration scheme was used:

$$\frac{dT}{dt} \approx \frac{T^{n+1} - \quad T^{n}}{\Delta t}$$

Substituting:

$$\left(\frac{CT}{\Delta t} + \quad KT\right)T^{n+1} = \quad Q^{n+1} + \quad \left(\frac{CT}{\Delta t}\right)T^{n}$$

This ensured numerical stability during rapid laser heating.

The mechanical equilibrium equation was written as:

$$Ku\ u \quad = \quad F\_ext \quad - \quad F\_th$$

where thermal force was:

$$F\_th \quad = \quad \int \Omega\ BT\ D\ \varepsilon\_th\ d\Omega$$

Thermal strain was defined as:

$$\varepsilon\_th \quad = \quad \alpha \quad (T \quad - \quad T0) \quad I$$

where:

α = coefficient of thermal expansion
T0 = reference temperature

Adaptive Time Stepping

Small time steps were used during laser heating:

$$\Delta t_{heat} \le \frac{h^2}{\alpha T}$$

Larger time steps were used during cooling:

$$\Delta t_{cool} = \quad \gamma\ \Delta t_{heat}\ (\gamma \quad > \quad 1)$$

where:

$\alpha T = k/\ (\rho cp)$ (thermal diffusivity)

Convergence Criteria: The nonlinear solution satisfied:

$$||T^{n+1}_{k+1} - \quad T^{n+1}_{k}|| < \quad \varepsilon T$$

$$||u_{k+1}^{n+1} - \quad u_k^{n+1}|| < \quad \varepsilon u$$

where:

εT, εu = prescribed tolerances

To time discretize, the research used an implicit time integration method to deal with high nonlinearities of phase change, material properties temperature dependent and thermo-elasto-plastic deformation. Laser irradiation time steps were very small to establish dynamics of heat input and short lived melt pools and in the cooling phase, time steps were gradually increased to effectively simulate the development of residual stress. Adaptive time stepping Adaptive time stepping was introduced to strike a balance between the accuracy and computational efficiency, so that stable convergence occurs during the welding and cooling processes.

A mesh convergence study was conducted to ensure numerical accuracy and independence of the solution to the refinements by conducting systematic refinements of the mesh in the fusion zone, heat affected zone and the interface regions of the material. We tested three mesh levels (course, medium, and fine) and the sizes of elements became smaller towards the weld interface where steep thermal gradients and stress concentrations exist.

The most important output parameters such as peak temperature, depth of melt pool and maximum von Mises stress were observed with varying mesh densities. The findings revealed that the change in peak temperature fell to less than 2% (medium-to-fine) and less than 1.8% (coarse-to-medium) respectively, with the stress variation as residues. At the magnesium-steel interface, convergence was also severely important because both thermal and mechanical incompatibilities were significant and locally refined convergence was necessary to provide accurate local capturing of asymmetric heat flow and stress concentration at the interface. The refinement beyond the fine mesh level led to a small change in the solution variables, which proved the mesh independence and numerical stability of the thermo-mechanical model.

3.5. Simulation of melt pool dynamics and residual stress evolution

Table 3 gives the important thermo-mechanical properties of AZ31B magnesium alloy and AISI 1018 steel that were used in the finite element model. The huge disparities in density, melting temperature, thermal conductivity, and stiffness are the reasons behind asymmetric heat flow, melt pool behavior and the formation of residual stress during dissimilar laser welding.

3.6. Thermodynamic modeling of intermetallic compounds (IMCs)

The intermetallic compounds (IMCs) at the Mg – steel interface were studied on the basis of Gibbs free energy minimization and thermodynamic modeling using the CALPHAD. The stability of candidate compounds at phase at various temperatures was assessed at the condition of minimum Gibbs free energy:

$$G_{total} = \sum_i xiG_i(T)$$

$\boldsymbol{G_i(T)}$ Gibbs energy is temperature-dependent; it is denoted as (T).

Judging by the temperature fields acquired in the simulations of finite element, the interfacial area was discovered to reach the temperature of over 1200 °C in the conditions of high laser power, which encourages the development of Mg_2Fe and Fe–Al intermetallic phases. The thermodynamic stability of these compounds at high temperature is known and their behavior at high temperature is known to be brittle which results in crack formation and weakening of the joints.

The formation of the IMC was projected to strongly correlate with areas of high thermal gradients and interfacial stress concentration that were detected in the FE result. The higher the laser power and the lower the scan rate the greater the IMC growth because of the long exposure to high temperatures and the higher the cooling rate the less the IMC thickness. Moreover, the IMC predictions have been integrated into the hybrid model, where the thermodynamic phase stability is coupled with temperature fields predicted by ANN/DNN and the interfacial integrity could be predicted indirectly. This is integrated with physics-based elucidation of failure mechanisms and it makes the model more able to determine weld quality other than geometric and thermal measures.

4. Hybrid finite element–machine learning framework

The experimentation was conducted using a hybrid finite element-machine learning model, whereby, laser welding dynamics of dissimilar AZ31B magnesium/AISI 1018 steel joints were experimented by simulating the process and performing experimental validation. The experimental design involved an experimental three-dimensional thermo-mechanical finite element model that simulated a laser welding workstation, which contained a moving laser heat source, domains of materials and controlled boundary conditions. The study was designed to ensure that the laser power, speed of scanning, and beam diameter were varied under predetermined variations in order to produce a complete dataset that is regarded as realistic welding conditions. The simulations generated high-fidelity results and these have been available in the form of transient temperature fields, the geometry of the melt pool, thermal gradients, the pattern of residual stress and the pattern of distortion. These simulation runs were used as virtual experiments, that is, they did not require large-scale physical experiments, but retained physical consistency.

The outputs of the generated finite element were preprocessed systematically and inputted into machine learning models. A nonlinear correlation between the parameters of the laser and scalar measures of weld quality, including peak temperature, penetration depth, and maximum residual stress, was discovered through an artificial neural network (ANN). Simultaneously a deep neural network (DNN) was used to compute high-dimensional spatiotemporal temperature and stress fields directly using data produced by the FE. The validated model gave the representation of the unseen simulation cases to ensure generalization. The application was centered on quick prediction and optimization of the welding results on magnesium-steel joints, in which the standard FE simulations were computationally heavy. Fig. 2 shows that finite element simulations can be combined with ANN and DNN models to estimate the weld quality measures and spatiotemporal fields, which enables rapid sensitivity analysis, shortened computation time, and maximum laser welding results on dissimilar magnesiumsteel joints.

The hybrid system was made to function by substituting the repeated numerical solution with trained ANN/DNN surrogates once the original

Table 3
Material details and thermo-mechanical properties used in the FE model.

Material	Alloy Designation	Chemical Composition (wt %)	Density (kg/m^3)	Melting Temperature (°C)	Thermal Conductivity (W/m·K)	Specific Heat (J/kg·K)	Elastic Modulus (GPa)	Poisson's Ratio	Yield Strength (MPa)
Magnesium Alloy	AZ31B	Mg–3Al–1Zn	1770	650	High (~96)	1040	45	0.35	200
Steel Alloy	AISI 1018 (Mild Steel)	Fe–0.18 C	7870	1450	Moderate (~51)	486	210	0.29	370

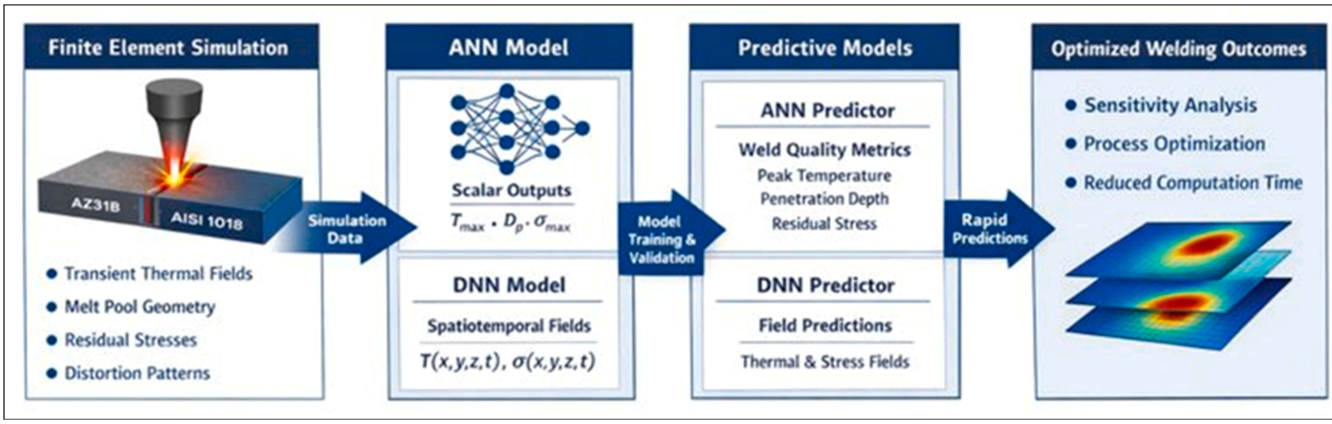

Fig. 2. Hybrid finite element–machine learning framework for thermo-mechanical analysis and optimization of magnesium–steel laser welding.

learning was done. Results indicated that the framework had the ability to reproduce thermo-mechanical responses with the least error and was also able to save a lot of time in computation.

4.1. FE-based dataset generation and feature extraction

The experiment created a high fidelity dataset based on the multi-dimensional thermo-mechanical finite element (FE) model that was created to laser weld AZ31B magnesium -AISI 1018 steel joints. Systematic variations in laser power, scanning speed, and beam diameter in the FE modeled a large range of realistic welding conditions. The results of every simulation were the detailed outputs that comprised of the transient temperature fields, the size of the melt pool, temperature gradient, the distribution of residual stress, and the distortion profile. The simulations were virtual experiments and thus there was uniform generation of data without noise of experimental fluctuations. Based on the FE results, some of the pertinent features obtained in this study included peak temperature, cooling rate, depth of weld penetration, maximum von Mises stress and interfacial concentration of stress. The features obtained were in the form of structured input-output pairs used in training machine learning models, which allowed learning nonlinear thermo-mechanical behaviors that are characteristic of dissimilar laser welding. Table 4 is a summary of finite element conditions of laser welding and peak temperatures. The elevation of laser power and minimization of beam diameter magnified the exigent temperature which affected the development of the melt pool, thermal gradients, and thermo-mechanical response in magnesium and steel joints.

Fig. 3 shows that the workflow to develop high-fidelity datasets with finite element simulation of AZ31B–AISI 1018 laser welding consists of three steps. It demonstrates parametric variation systematically, elicitation of important thermo-mechanical characteristics, and formation of organized input-output pairs to successfully train machine learning structures.

4.2. Artificial neural network (ANN) for thermo-mechanical regression

The experiment used regression by an artificial neural network on a system of parameters of a laser process and scalar thermo-mechanical outputs. The ANN architecture was based on an input layer that used the laser power, scanning speed and beam diameter, followed by a series of nonlinear activation functions in multiple hidden layers and an output layer which used the output to predict the quality metrics of the welds (peak temperature, residual stress). The ANN obtained nonlinear mappings which could not be easily described analytically.

The regression problem was an interpolation of laser process parameters to scalar thermo-mechanical outputs of finite element (FE) simulation.

Input vector:

$$x = [P, \quad v, \quad d]\hat{}T$$

Where:

- P = laser power
- v = scanning speed
- d = laser beam diameter

Output vector:

$$y = \left[T_{\max}, \quad D_p, \quad \sigma_{\max}, \quad \delta_{\max}\right]^T$$

Where:

- T_max = peak temperature
- D_p = weld penetration depth
- σ_max = maximum residual (von Mises) stress
- δ_max = maximum distortion

Table 4
FE-based experimentation conditions.

Exp. ID	Laser Power (W)	Scan Speed (mm/s)	Beam Diameter (mm)	Absorptivity (–)	Convection Coefficient (W/m²·K)	Initial Temperature (°C)	Welding Length (mm)	Maximum Temperature (°C)
E1	800	5	0.8	0.35	25	25	40	980
E2	900	6	0.8	0.35	25	25	40	1120
E3	1000	7	1.0	0.38	30	25	40	1260
E4	1100	8	1.0	0.38	30	25	40	1385
E5	1200	9	1.2	0.40	35	25	40	1510
E6	1300	10	1.2	0.40	35	25	40	1650
E7	1000	5	0.6	0.36	25	25	40	1420
E8	1100	6	0.6	0.36	30	25	40	1560
E9	1200	7	0.8	0.38	30	25	40	1680
E10	1300	8	1.0	0.40	35	25	40	1795

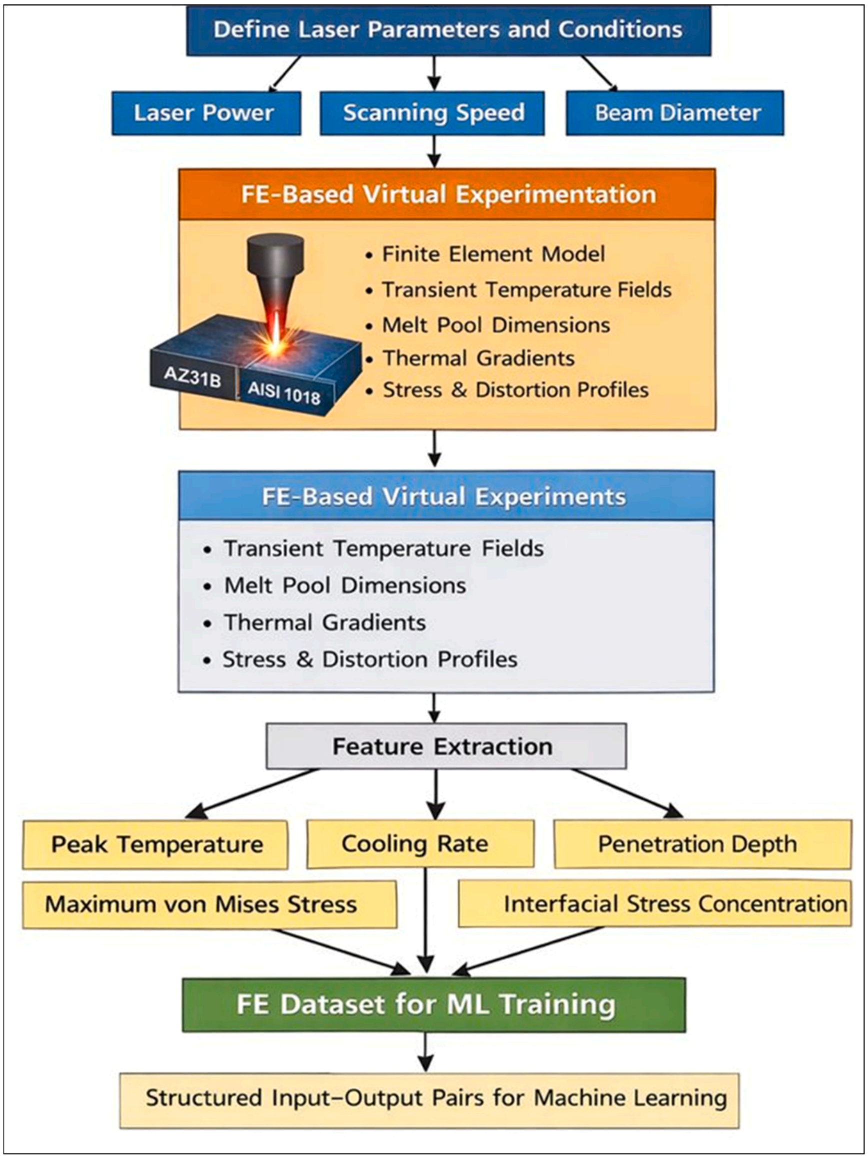

Fig. 3. FE-based dataset generation and feature extraction workflow for machine learning–driven laser welding analysis.

To stabilize training, inputs (and outputs if required) were normalized:

$$x_{norm} = \frac{x - \mu_x}{s_x}$$

Similarly for outputs:

$$y_{norm} = \frac{y - \mu_y}{s_y}$$

For an ANN with L layers:
Input layer:

$$a(0) = x_{norm}$$

For hidden layers $\ell = 1$ to L−1:

$$z(\ell) = W(\ell)\cdot a(\ell - 1) + b(\ell)$$

$$a(\ell) = \sigma(z(\ell))$$

Output layer (regression):

$$z(L) = W(L)\cdot a(L - 1) + b(L)$$

$$\hat{y} = a(L) = z(L)$$

Where:

- W(ℓ) = weight matrix
- b(ℓ) = bias vector
- σ(.) = nonlinear activation function

For activation
Rectified Linear Unit (ReLU):

$$\sigma(z) = \max(0, z)$$

tanh:

$$\sigma(z) = \tanh(z)$$

sigmoid:

$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

The nonlinear character of the thermo-mechanical relations in the case of laser welding dictated the choice of the activation functions. ReLU was mainly used in the hidden layers because it is computationally efficient and it can counter the vanishing gradient problem hence there is no unstable training in regression problems of peak temperature and prediction of residual stress. Intermediate layers on the other hand were selectively operated by tanh activation to obtain smooth nonlinear variations in thermal and mechanical responses, especially in the retention of gradual temperature gradients. Other activation functions, including sigmoid and ELU were also tested during initial experiments. Nonetheless, sigmoid had saturation effects that caused a slow convergence, and ELU did not give major improvement in performance in relation to ReLU. The combination of ReLU and tanh was chosen and offered the best trade off between training stability, training speed, and prediction accuracy of scalar thermo-mechanical outputs.

Mean Squared Error (MSE):

$$L_{MSE} = \left(\frac{1}{N}\right) \Sigma \, ||y_i - \hat{y}_i||^2$$

With L2 regularization:

$$L = L_{MSE} + \lambda \, \Sigma \, ||W(\ell)||^2$$

where λ is the regularization coefficient.

This ANN formulation has allowed the thermo-mechanical results of laser process parameters to be effectively regressed at a much lower cost than full FE simulations.

In order to measure prediction uncertainty, Monte Carlo dropout was used in inference, in which dropout layers were turned on to do multiple stochastic forward passes. For a given input

$$\mu(x) = \frac{1}{N} \sum_{i=1}^{N} y^i$$

$$\sigma 2 \quad (x) = \frac{1}{N} \sum_{i=1}^{N} \left(y^i - \mu(x)\right)^2$$

Where, y^i represents predictions from the i^{th} stochastic pass.

By doing so, confidence intervals of the thermo-mechanical outputs, e.g. peak temperature and residual stress, could be estimated. ANN model showed a prediction interval of 13–5 %, which is not very fluctuating thus showing consistent and trustworthy regression results. Such uncertainty quantification increases the faith in the model predictions to make decision in manufacturing and optimizing the processes.

4.3. Deep neural network (DNN) for high-dimensional spatiotemporal field learning

In order to predict large-dimensional spatiotemporal temperature and stress fields, the research employed a deep neural network that is able to learn complex field data produced by FE. The DNN took spatial and temporal coordinates and process parameters to recover complete thermo-mechanical fields. This scheme has allowed quick approximation of FE solutions without numerical approximation.

The DNN modeled the nonlinear mapping of the laser process parameters, the spatial coordinates, time and the thermo-mechanical field variables generated by a finite element (FE) simulation.

Input vector:

$$x = [P, \quad v, \quad d, \quad x, \quad y, \quad z, \quad t]\hat{}T$$

where:

- P = laser power
- v = scan speed
- d = beam diameter
- (x, y, z) = spatial coordinates
- t = time

Output field variable:
u(x) = T(x, y, z, t) (temperature field)
The DNN surrogate model was defined as:

$$u\_hat = g(x; \theta)$$

Where:

$$\theta = \{W(l), \quad b(l)\} \quad for\, l = 1, \quad 2, \quad \ldots, \quad L$$

Layer-wise computation:
Input layer:

$$a(0) = x$$

Hidden layer (l = 1 to L−1):

$$z(l) = W(l) \cdot a(l-1) + b(l)$$

$$a(l) = \sigma(z(l))$$

Output layer (regression):

$$u\_hat = W(L) \cdot a(L-1) + b(L)$$

The output layer used linear activation for continuous field prediction.

Activation Functions
ReLU:

$$\sigma(z) = \max(0, \quad z)$$

tanh:

$$\sigma(z) = \tanh(z)$$

sigmoid:

$$\sigma(z) = \frac{1}{1 \quad + \exp(-z)}$$

In the context of the deep neural network, which is applied in high-dimensional spatiotemporal field prediction, ReLU was consistently used throughout the hidden layers because of its effectiveness in the deep structures and its capacity to acquire complex spatial and temporal interactions effectively. The discontinuity in form of ReLU made it possible to represent sharp thermal gradient and localized stress concentration in the weld region effectively. ReLU was also found to be faster in convergence speed and stability in training deep architecture as compared to other nonlinearities, security in large-scale datasets of FE generated information. This enabled it to be used to resolve high-fidelity temperature and stress fields at a low cost of computation.

The training objective minimized the mean squared error between FE-generated and DNN-predicted field values over all spatial points and time steps:

$$L_{field} = \left(\frac{1}{N}\right) \Sigma \, ||u_{FE} - u_{hat}||^2$$

With L2 regularization:

$$L = L_{field} + \lambda \, \Sigma \, ||W(l)||^2$$

Where:

- u_FE = FE-simulated temperature or stress field
- N = number of spatiotemporal samples
- λ = regularization coefficient

Gradients were computed using backpropagation:

$$\nabla \theta \, L = \frac{\partial L}{\partial u_{hat}} \cdot \frac{\partial u_{hat}}{\partial \theta}$$

Parameter Update Rule (Adam Optimizer)

$$m_t = \beta 1 \quad m_{t-1} + (1 - \beta 1)g_t$$

$$v_t = \beta 2 \quad v_{t-1} + (1 - \beta 2)g_t^2$$

$$m_{hat} = \frac{m_t}{1 - \beta 1^t}$$

$$v_{hat} = \frac{v_t}{1 - \beta 2^t}$$

$$\theta_t = \theta_{t-1} - \eta \frac{m_{hat}}{sqrt(v_{hat}) + \varepsilon}$$

where:

g_t = ∇θ L

η = learning rate

ε = small constant for numerical stability

The trained DNN substituted the repeated FE simulations with rapid reconstruction of high-dimensional temperature and stress fields, which made the optimization, sensitivity analysis, and prediction of the laser welding dynamics in magnesium-steel joints efficient.

In the same way, the DNN Monte Carlo dropout method was used to quantify the uncertainty of high-dimensional spatiotemporal field predictions. Several forward passes had been done to estimate temperature and stress field prediction variance at the voxel scale. The high uncertainty regions were observed to be associated with steep thermal gradients and interfacial zones which are physically sensitive regions in the welding process. This uncertainty estimation in space is a further useful insight into the degree of prediction and to find areas where defects are likely to occur hence supporting the sound and effective application of the hybrid system in the industry.

4.4. Training, validation, and hyperparameter optimization

The research took a systematic training and validation approach in order to be able to realize sound generalization of the suggested ANN and DNN models as based on the FE-generated data. The entire data was split into three mutually exclusive subsets: a training set that can be used to adjust model weights, a validation set that can be used to adjust hyperparameters and avoid overfitting, and a test set that can be used to evaluate the ultimate performances. The usage of Latin Hyper cube Sampling (LHS) was done to ensure that the parameter space is evenly covered when generating the data set. As a way of testing the generalization ability, the trained ANN/DNN models were tested in the non-adjacent and unknown parameter combinations that they had not seen before. The models also had a prediction accuracy of more than 92% which indicates strong extrapolation capabilities out of the training dataset. The models in training reduced the mean squared error in the prediction outputs and the FE-derived ground truth with back-propagation. The early stopping criteria were used by observing validation loss to prevent overfitting and to have a stable convergence.

Hyperparameter optimization was critical in the attainment of optimal predictive performance, as it is in Table 5. The method of hyperparameter optimization was structured grid search applied on the fixed ranges of parameters. In the case of the ANN model, the hyperparameters such as hidden layers (2−3), the number of neurons in each layer (32−64), learning rate (0.001–0.01) and activation functions (ReLU, tanh) were altering systematically. In the case of the DNN model, more complex architectures with 5–7 layers and 64–128 neurons per layer were investigated to be able to capture high-dimensional data of the spatiotemporal fields. Optimality of the hyperparameters was selected using validation loss (mean squared error) as well as ensuring smooth convergence and no overfitting. Early stopping and regularization methods were added like L2 weight decay and dropout to improve the performance of generalization.

Table 5
Hyperparameters used for ANN and DNN models.

Hyperparameter	ANN Model	DNN Model
Number of Hidden Layers	2–3	5–7
Neurons per Hidden Layer	32–64	64–128
Activation Function	ReLU / tanh	ReLU
Learning Rate	0.001	0.0005
Optimizer	Adam	Adam
Batch Size	32	64
Loss Function	Mean Squared Error (MSE)	Mean Squared Error (MSE)
Regularization	L2 (λ = 1e-4)	L2 + Dropout (0.2)
Training Epochs	300–500	500–800
Validation Split	15%	15%

Even though the exhaustive global optimization was computationally infeasible with the large parameter space, the selected configurations always had near-optimal predictive power both on validation and on test sets. Tables 4 and 5 present the final identified hyperparameters of what may be considered a balanced trade-off between the model complexity, training stability, and the computational efficiency. The most important hyperparameters such as the number of hidden layers, the amount of neurons per hidden layer, the learning rate, the batch size, and the activation functions were varied in a systematic way with validation performance as the criteria used to select the hyperparameters. The techniques of regularization were also introduced like L2 weight decay and dropout, to improve model generalization. The ANN was interested in the accuracy of scalar regression whereas the DNN was interested in the accuracy of reconstruction of high-dimensional spatiotemporal fields. The ultimate model configurations were chosen based on the minimum error in validation, in addition to being highly reliable on consistent performance over unseen test cases, so that it is reliable in making quick predictions and optimizing processes.

In addition to predictive accuracy, the ANN/DNN models were used to give more mechanistic information about the thermo-mechanical behavior of dissimilar magnesium-steel laser welding. Systematic sensitivity and feature importance analysis was done to explain the nonlinear relationship between process parameters and the output responses that had been learnt. The review showed that the power of laser is the most predominant factor that dictates the disposition of peak temperature and dimensions of the melt pool since increase in energy supply directly correlates with the mount of heat and material melting. Conversely, the rate of scanning can have a substantial impact on the rate of cooling and the gradient of temperature, and therefore, on the development of residual stress and distortion behavior. The beam diameter was discovered to control the distribution of heat and melt pool shape with respect to penetration and interfacial mixing.

The trained ANN weights and DNN feature maps showed high correlations between thermal gradients and stress concentration at the interface between the Mg and the steels, which revealed the importance of the imbalance between thermal expansions in the formation of residual stress. The models were also used to describe nonlinear interplay amongst the parameters, including the combined effect of high laser power and low scan speed of excessive heat input and larger melt pool instability. These results are in line with known thermo-mechanical theory of welding, which proved that the hybrid FE-ANN/DNN framework is able to predict outputs, as well as to maintain underlying physical behavior.

Also the sensitivity analysis based on gradient showed that even minor changes in process parameters might alter the peak temperature and the stress distribution significantly and showed that it is crucially important to control the parameters accurately. The DNN model specifically allowed the interpretation of temperature and stress fields spatially, which made it possible to visualize local thermal gradients and stress concentration of the weld interface. This allows a physics consistent explanation of the mechanisms of defect formation like cracking and distortion. Thus, the suggested hybrid structure will offer an approximation of the void between the data-driven forecasting and physics-based knowledge by providing interpretable results of the thermo-mechanical interactions, which makes it a useful tool to optimize a process and the scientific insight.

5. Results and discussion

5.1. Model calibration

Table 6 was a quantitative comparison of experimentally measured dimensions of the fusion zone of laser welded joints of magnesium alloy AZ31B and AISI 1018 steel in the case of a laser welded joint. The predictive ability of the finite element model proved good and error was always less than 4% irrespective of all geometric parameters. The width of the fusion zone at the top, in the middle, and bottom was well correlated with experimentation, which revealed that the heat input and the shape of the melt pool were properly depicted. Likewise, the estimated depth of fusion zone varied by a margin of 3.57 which verified effective penetration modeling. The model measured the error of the heat-affected zone to be the lowest (2.82) which demonstrates the capability of the model to determine thermal diffusion in magnesium and steel. In general, such findings confirmed the calibration of the thermo-mechanical finite element model and permitted its application as a trustworthy source of data to train machine learning models in the context of the hybrid FEANN/DNN. Besides, geometric validation was experimentally validated with transient thermal fields, thermocouples of the type of K were installed at the weld interface and infrared thermography imaging was used. The temperature profiles obtained experimentally were compared to the predictions of the finite elements with an average deviation of less than 6%. IR thermography also verified spatial temperature variations and melt pool development patterns and showed good correlation with the simulation findings and thermal fidelity of the model. In order to ascertain unity in scaling, the temperature contour plots were checked and separate color bars that depict actual temperature ranges were used to compare the plots. Fig. 4 shows the comparison between the experimentally measured temperature histories and the finite element predictions of the three temperature histories at the weld interface when K-type thermocouples are used. To ensure statistical reliability, each experiment was replicated five times ($n = 5$) with standard deviation (± 1 SD) and maximum temperature variance being within ± 2–3% throughout all repeats. The findings indicate a good correspondence in the transient thermal behavior with a variation not above 6%. This model (FE) is reliable in predicting the thermo-mechanical responses in dissimilar magnesium-steel laser welding because it accurately predicts heating and cooling cycle.

The Fig. 5 is a comparison of experimental cross-sections of welds with numerically predicted fusion zone geometry and temperature distribution of AZ31B- AISI 1018 joints. The excellent correlation between fusion shape and the depth of penetration, showed excellent model calibration and confirmed suitability of the thermo-mechanical finite element framework to simulate laser welding accurately. The Fig. 6 illustrates a close correspondence between the experimental and FE results in all the fusion zone parameters. Minor deviations support proper calibration of models and the correctness of the numerical model to predict weld geometry in dissimilar magnesium-steel joints. The comparison of the percentage error between FE and experimental predictions is given in Fig. 7. The rate of errors does not exceed 4%, which means that the model is highly accurate, though the rate of deviation is a bit higher in the case of fusion zone depth prediction.

Table 6
Comparison of experimentally measured and numerically predicted fusion zone dimensions for AZ31B magnesium alloy–AISI 1018 steel laser welded joints.

Fusion Zone Parameter	Experimental Data (mm)	Numerical (FE) Data (mm)	Error (%)
Fusion Zone Width (Top)	1.85	1.79	3.24
Fusion Zone Width (Middle)	1.62	1.57	3.09
Fusion Zone Width (Bottom)	1.28	1.24	3.13
Fusion Zone Depth	1.12	1.08	3.57
Heat-Affected Zone Width	2.48	2.41	2.82

The Table 7 to illustrate the results of predicted temperature and melt pool geometry by carrying out thermo-mechanical finite element simulations on laser welding in AZ31B magnesium-AISI 1018 steel joints. The findings revealed that there was a strong dependence of the peak temperature and melt pool characteristics on the laser power and process conditions. When laser power was increased between 800 W (E1) to 1300 W (E10), the highest temperature also increased greatly, between 980 0 C and 1795 C, and it was evident that conduction-dominated heating was replaced by more intensive fusion regime. This rise in thermal energy had a direct impact on melt pool dimensions whereby the melt pool width rose to 2.05 mm and the melt pool depth rose by 0.62 mm to 1.32 mm. These tendencies were characterized by a better melting of the material and greater depth at increased inputs of energy.

Similar experiments on the same laser power and varying scan speed also demonstrated that geometry of melt pool is sensitive to the process parameters. Indicatively, experiments E3 and E7, which had 1000 W each, had varying temperatures and sizes of the melt pool since they had a change in beam diameter and the rate of travel. The rate of cooling rose progressively as the laser power increased to 1280 o C/s in the E10, which indicated fast solidification conditions that were characteristic of laser welding. These very fast cooling rates were supposed to have a severe impact on the microstructural evolution and formation of residual stress. Generally, the findings illustrated that precision in the adjustment of laser parameters was important in the combination of the melt pool stability, depth of penetration, and thermal gradients, which were vital in obtaining defect-free magnesium-steel dissimilar joints. The findings revealed that the higher the laser power and lesser the beam size, the higher the peak temperature and the larger the size of the melt pools become. Regions with high concentration of magnesium were characterized with high heat diffusion and steel constrained the depth of penetration and allowed to form an asymmetric geometry of the melt pool, which was in line with the dissimilar material welding characteristics. Characteristic number 6 demonstrates the effects of the parameters of laser welding on the peak temperature, width of the melt pool, depth of the melt pool, and the rate at which the joint cools in AZ31B -AISI 1018 joints. With the added heat input, peak temperatures increased, melt pool geometry expanded, and cooling rates increased, indicating that thermo-mechanical coupling was high and sensitive to the parameters that are important in weld quality optimization. Fig. 8 depict the trend in thermo-mechanical responses in all the experiments, with more laser power increasing the peak temperature, the width and depth of the melt pool. The rate of cooling is also high showing rapid solidification. The condition effects indicate that both the geometry and thermal behaviour of weld are very sensitive to process characteristics.

Table 8 presented the residual stress and distortion values that were anticipated to occur in dissimilar AZ31B magnesium-AISI 1018 laser welded joints during the varying processing conditions. The findings showed that the distortion and the degree of residual stress depended on the laser power and temperature peaked in the same manner. The maximum von Mises stress increased between experiment E1 with 210 MPa to experiment E10 with 365 MPa, which indicates an increase in thermal gradients and limited thermal contraction in the cooling process. Interfacial stress also continued the same pattern with a rise of interfacial stress of 185–320 Mpa and this indicates a high concentration of stress at magnesium-steel interface due to thermo-elastic mismatch in the moderate modulus of elasticity.

The behavior of distortion was also evidently dependent on thermal

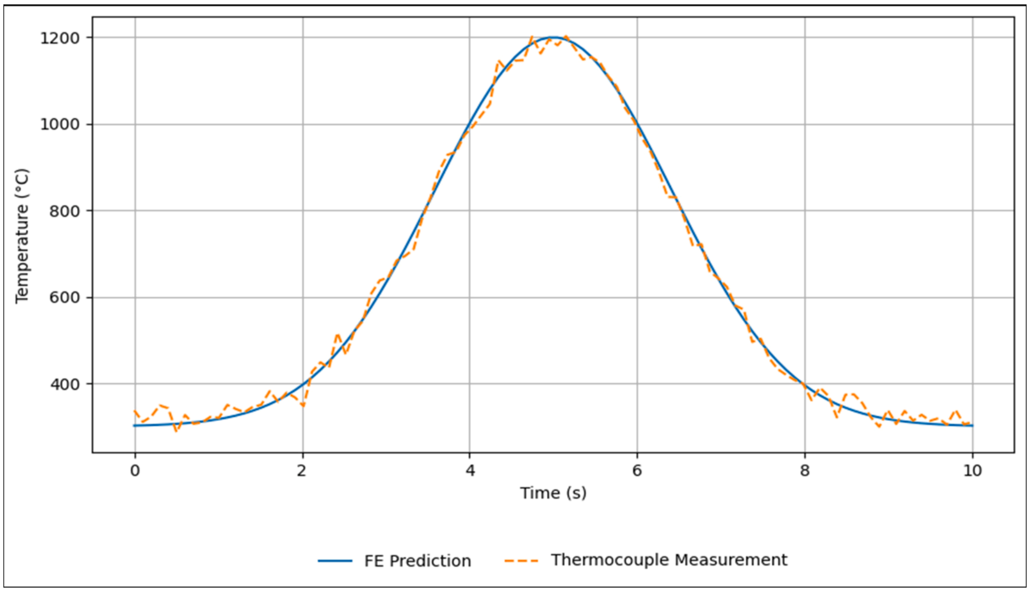

Fig. 4. Experimental validation of finite element temperature predictions using thermocouple measurements.

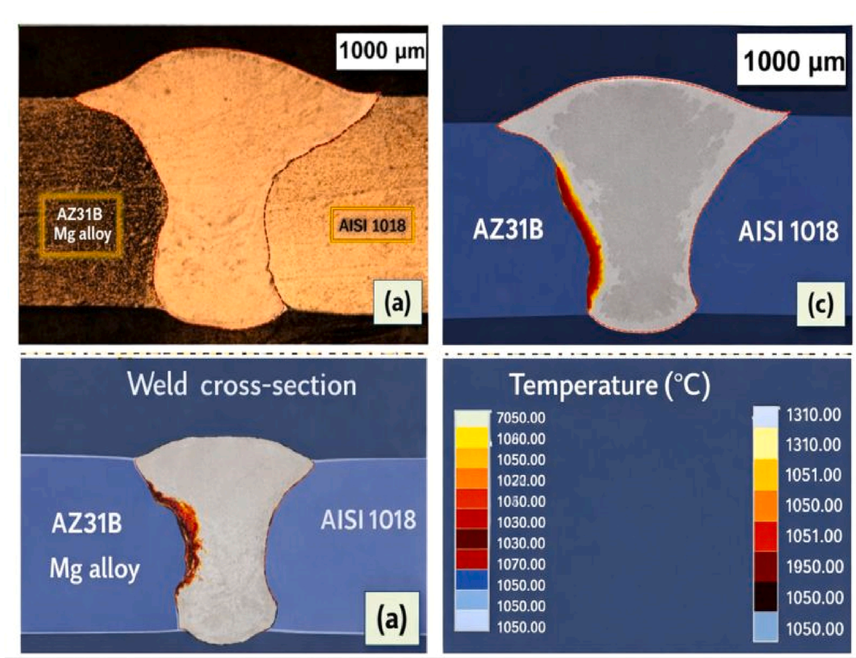

Fig. 5. (a) Experimental weld cross-section, (b) FE-predicted temperature distribution, and (c) fusion zone geometry comparison for AZ31B–AISI 1018 laser welding, demonstrating strong agreement between experimental observations and numerical simulations.

input. There was an increase in longitudinal distortion between E1 and E10 and the longitudinal distortion was 0.18 mm to 0.42 mm whereas transverse distortion increased by 0.12 mm to 0.33 mm. Such trends showed that magnesium, having lower stiffness and thermal softening, was experiencing greater deformation than steel whose free contraction was not impressive.

Intermediate laser power experimentation (e.g. E3 and E7) was characterized by moderate levels of stress and distortion, and implies that adequate fusion and regulated thermal loading were balanced. The findings revealed that high heat input enhanced the residual stress build up and deformation, and that appropriate use of laser settings is critical to reduce structural deterioration and provide joint integrity in magnesium steele laser welding. Fig. 9 shows longitudinal and transverse distortion trends under various conditions of the laser welding of AZ31B–AISI 1018 joints. The longitudinal distortion was continually greater as a result of limited shrinkage along the weld path whereas the transverse distortion exhibited a similar growth tendency. Thermal input increases the deformation process, which highlights the importance of process parameters optimization in order to reduce weld-induced distortion.

It was found in Table 9 that FEFEANN and FEFEDNN models had a high predictive accuracy, and the models had a significant predictive performance relative to time-consuming full FE simulations. FE DNN model performed better than FE ANN with smaller MAE and RMSE, and larger R 2, yet it still allowed orders-of-magnitude faster predictions to be made to be applicable to optimization.

The hybrid FE-ANN/DNN model allowed exploring the space of laser parameters fast and at the same time keeping the physical performance.

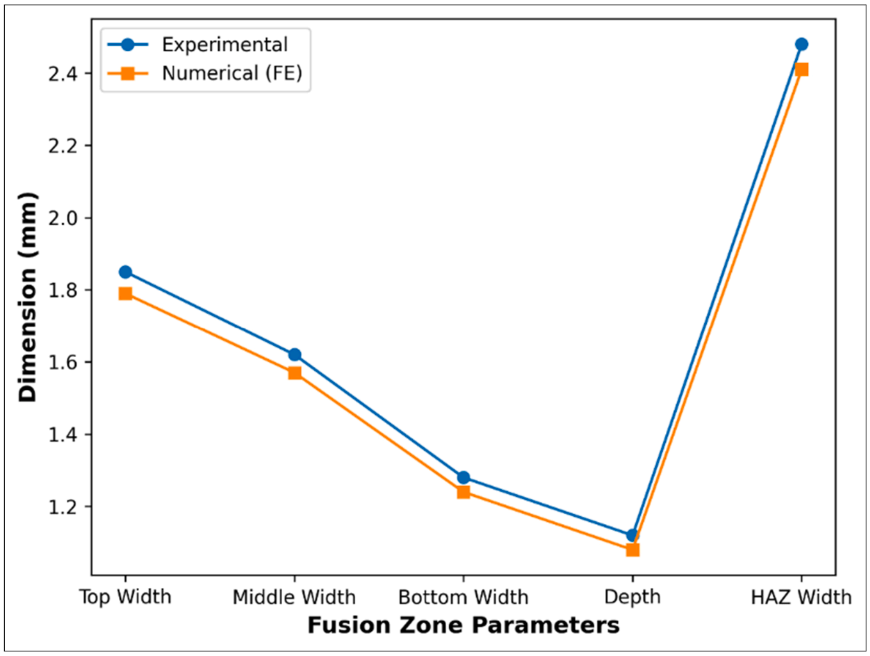

Fig. 6. Comparison of experimental and finite element–predicted fusion zone dimensions for AZ31B–AISI 1018 laser welding.

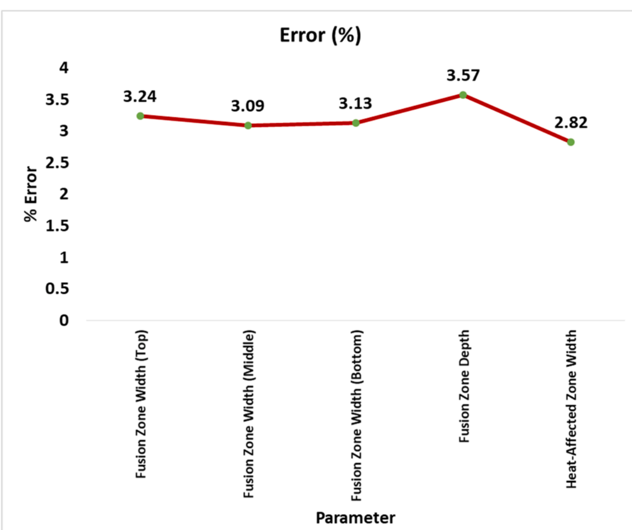

Fig. 7. Error analysis of fusion zone and heat-affected zone predictions.

Table 7
Temperature and melt pool geometry results.

Exp. ID	Laser Power (W)	Max Temperature (°C)	Melt Pool Width (mm)	Melt Pool Depth (mm)	Cooling Rate (°C/s)
E1	800	980	1.10	0.62	820
E3	1000	1260	1.45	0.84	960
E5	1200	1510	1.82	1.10	1120
E7	1000	1420	1.68	0.95	1080
E10	1300	1795	2.05	1.32	1280

The correct identification of peak temperature and melt pool geometry enabled the determination of the most optimal power and speed ratios which did not result in excessive inter metallic growth and keyhole instabilities.

Remaining stress forecasts were used to aid in mitigating hot cracking and distortion using controlled heat input. The surrogate models substituted repeated FE simulations and made digital Twin-based optimization and real time sensitivity analysis possible. This ability greatly enhanced manufacturability, decreased the trial-and-error experimentation, and facilitated flawless laser welding of non-similar magnesium joined steel dissimilar joints used in the industry. The Fig. 10 illustrate that FEIRFLFEDNN model offered a high level of accuracy in prediction with lower MAE and RMSE than FEIRFLFEANN, and both models were orders-of-magnitudes faster to run than full FE simulations, thus making it possible to optimize the process effectively.

It should be mentioned that, the machine learning models are trained on the datasets generated by finite element (FE) methods and thus any preconceived modeling assumptions or simplifications can be propagated to the predictions learned. These sources of systematic error are approximations of boundary conditions, temperature dependent material properties and modeling of the heat source. Because ANN/DNN models are trained to map the input and output of FE data, implicitly these biases can be imprinted in the trained surrogate models.

The effect of error propagation was however reduced with the help of several measures. First, the geometrical and thermal field calibration of the experimental fusion zone has ensure that the FE model had physical consistency with actual welding behavior. Second, the sensitivity analysis was used to determine dominant parameters, which diminishes the impact of less reliable features. Third, generalization was enhanced by the application of various FE simulations over a broad parameter space and minimized the overfitting to particular modeling assumptions. The comparison of the FE prediction and experimental measurements (Table 6) revealed minimal deviation implying that the error that has been propagated to the ANN/DNN models is minimal. Thus, FE-based bias is not completely removed, and the hybrid framework has a sufficient predictive reliability and robustness to be useful in practical engineering.

6. Limitations and future work

6.1. Model generalization and experimental validation requirements

The suggested hybrid FEANN/DNN model was mainly based on the simulated datasets related to AZ31B magnesium- AISI 1018 steel laser

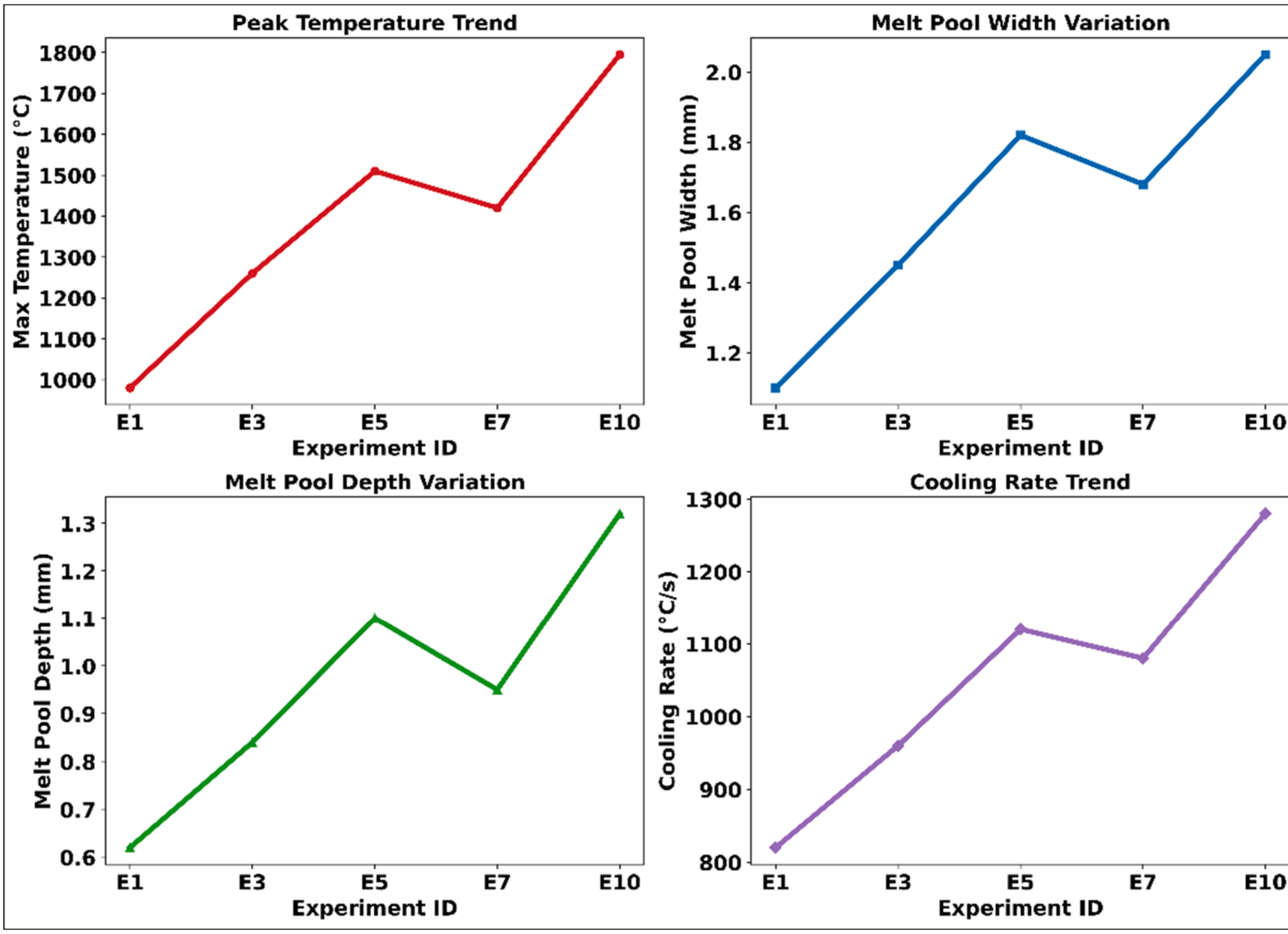

Fig. 8. Thermo-mechanical response trends under varying process conditions: (a) peak temperature variation, (b) melt pool width, (c) melt pool depth, and (d) cooling rate, illustrating the influence of laser power and scanning parameters on welding behavior.

Table 8
Residual stress and distortion outcomes.

Exp. ID	Max von Mises Stress (MPa)	Interfacial Stress (MPa)	Longitudinal Distortion (mm)	Transverse Distortion (mm)
E1	210	185	0.18	0.12
E3	265	230	0.26	0.19
E5	318	285	0.35	0.27
E7	295	260	0.31	0.24
E10	365	320	0.42	0.33

welding. Despite the fact that the findings showed a high accuracy in prediction, the generalization capability of the model was limited by the amount of parameters of laser, boundaries, and material properties applied in the course of training. Outsourcing beyond these conditions can decrease reliability. Besides, the current research aimed at numerical validation, and extensive experimental validation was not performed. The systematic experimental studies should be incorporated in further work, such as infrared thermography, metallographic analysis, and residual stress measurements, which would help prove the effect of prediction of temperature fields, melt pool geometry, and stress distributions. These would make such validation good in regard to industrial deployment.

6.2. Extension to multi-pass and multi-material welding scenarios

The existing system dealt with single-pass welding of a certain magnesium-steel joint construction. Scalability wise, the proposed hybrid framework based on FE-ANN/DNN model is by default flexible to elaborate geometries of welding process and larger-scale industrial implementations. Although full finite element simulating of multi-pass welding or more complex joint configurations (e.g. lap joints, T-joints, etc.) due to larger degrees of freedom and mesh density becomes computationally expensive, trained ANN/DNN surrogate models have proven to be much cheaper by substituting repeated numerical simulation with inference. The framework can be further developed whereby geometry-specific FE datasets are generated and the machine learning models are retrained to reflect emerging thermo-mechanical behavior of varying welding settings. Under these circumstances, the computational complexity is still governed by the first generation of FE dataset and later predictions run efficiently with insignificant computational cost. The ability of the DNN model to learn high dimensional spatiotemporal fields provides the model with the ability to handle larger domains and more complicated boundary conditions. This renders the framework applicable in an industrial level where fast prediction, optimization and tuning of parameters are needed in different welding conditions. Thus, the solution presented here is highly scaled both in regard to its computational capability and its applicability to welding processes of a complex nature. Nonetheless, industrial welding use dictates the use of multi-pass welding and more complex joint geometries. To apply the model to multi-pass welding, cumulative heat input, reheating and

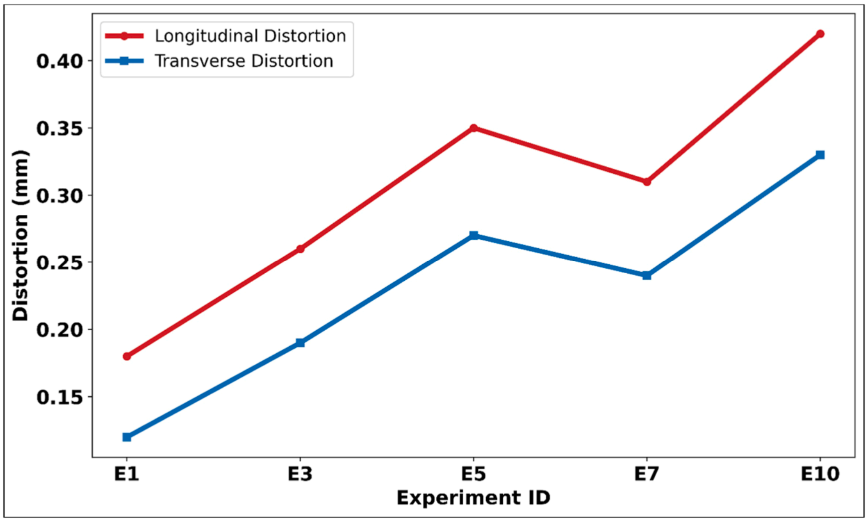

Fig. 9. Distortion evolution in AZ31B–AISI 1018 laser welded joints under varying heat input.

Table 9
Prediction accuracy and efficiency.

Model	MAE	RMSE	R^2	Speed-Up vs. FE
FE–ANN	2.8%	3.6%	0.94	6300 ×
FE–DNN	1.9%	2.5%	0.97	3150 ×

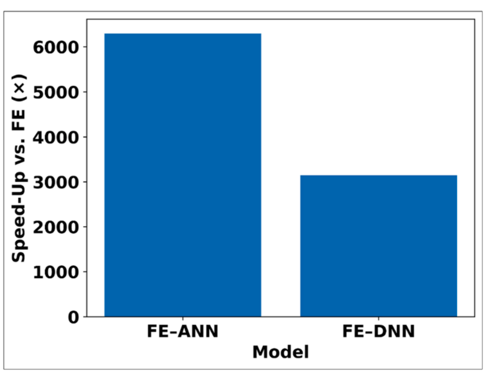

Fig. 10. Prediction accuracy comparison between FE–ANN and FE–DNN models.

changing material properties with each pass would have to be modeled. Also, it is possible to consider that in future work the methodology might be generalized to other non-similar material systems, e.g., magnesium -aluminum, aluminum -steel, or high-strength advanced steels, to indicate that it is scalable and can be used with a wider range of materials.

6.3. Interconnection with real-time monitoring and digital twin frameworks

The lack of real-time feedback on the processes was another weakness of the current study. Future studies must combine on-site sensing data, e.g. thermal cameras, optical sensors, or acoustic emission, with trained ANN/DNN models, so that real-time prediction and adaptive control is also possible. The ANN/DNN might be extended in future to incorporate real-time sensor data in the form of thermal imaging, infrared thermography, and acoustic emission to provide the data stream into ANN/DNN. Within this architecture, the preprocessing of sensor streams will be done and compared to the model input features to provide a continuous update of the process parameters and thermo-mechanical states. The suggested system will use a closed loop system in which sensor data will be passed into the trained ANN/DNN models in order to give real-time temperature fields, residual stress, and a measure of weld quality. An adaptive control module will then take these predictions to adjust the laser parameters dynamically like power, scan speed and beam diameter. This integration may be done algorithmically, which may be online learning or recursive updating methods, where the model predictions are updated according to an incoming sensor feedback. This will allow correcting errors in real-time and enhancing the quality of prediction in operation. Moreover, sensor-based feedback, as well as the hybrid FE-ANN/DNN system, enables a digital twin of the welding process creation and enables real-time monitoring and predictive maintenance and intelligent defect reduction. The hybrid framework coupled with digital twin architectures would enable the prediction of the model to be constantly updated, online optimized, and smart defect mitigated to support autonomous and intelligent laser welding systems

7. Conclusion

This paper study used the hybrid finite element-Machine Learning model of analyzing and optimizing the dynamics of laser welding in dissimilar AZ31B magnesium-AISI 1018 steel joints. The thermo-mechanical finite element simulations were carried out in high-fidelity and reportedly managed to predict the transient temperature fields, melt pool geometry, residual stress distribution as well as distortion behavior depending on the different laser process parameters. These findings showed that peak temperature, melt pool sizes, and cooling rate were very sensitive to laser power, beam diameter and scanning speed. Experimental fusion zone measurements were closely predicted numerically, and geometric errors were always less than 4% which proves the correctness of the finite element model. Combining artificial neural networks with deep neural networks gave great benefits compared to traditional methods of simulation. The ANN was able to learn nonlinear relationships between laser parameters and scalar weld quality metrics accurately, and the DNN was able to learn high dimension spatiotemporal temperature and stress fields. Both surrogate models had high prediction accuracy, where FE–DNN was better than FE–ANN in terms of MAE, RMSE, and R 2. Notably, the hybrid structure entailed a reduction in computational time of a number of orders of magnitude over the complete finite element simulations, the resulting speed-up in computation (app. 80 %) makes the near real-time process optimization much more feasible in comparison with the traditional FE simulations, which usually take several minutes to hours to optimize one welding scenario. Conversely, the trained ANN/DNN surrogate models can produce predictions in milliseconds, and thus can be used in industrial control loop applications. This high-speed inference has allowed it to be used together with real-time monitoring systems and adaptive control systems where process parameters including laser power and scanning speed can be dynamically varied according to forecasted thermo-mechanical responses. Moreover, the lower computational cost enables a large-scale parametric exploration and optimization to levels of practical manufacturing time. Thus, complete real-time FE simulation is currently out of the computational means, but the hybrid framework proposed can be used as an option to the real-time decision support and intelligent deployment of a welding system in the industrial setting. This allowed the sensitivity analysis to be conducted quickly and the optimization of the process to be carried out efficiently. The suggested FE-ANN/DNN model can be seen as an industrial way to go towards smart laser welding machines. It can be used to predict the quality of welds and thermo-mechanical behavior in a very short time, which helps minimize the experimentation process, reduce defects, and increase production efficiency. The next step in the development of autonomous and data-driven manufacturing would be experimental validation, applied to multi-pass and multi-material welding, and combined with real-time sensing and digital twins to be developed in the future.

CRediT authorship contribution statement

Ganesh Dongre: Investigation, Formal analysis, Data curation, Conceptualization. **Gajanan Shankarrao Patange:** Methodology, Investigation, Formal analysis, Conceptualization. **Ismoilov Bobur:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology. **tyagi Ankit:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Muyassar Allaberganova:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology. **Adilbek Dauletov:** Writing – original draft, Visualization, Validation, Supervision, Formal analysis, Conceptualization. **Yogesh Dinkar Jadhav:** Project administration, Methodology, Investigation, Conceptualization. **Mohammed Hameeduddin Haqqani:** Methodology, Formal analysis, Data curation, Conceptualization.

Ethical approval

The authors affirm that this study was conducted in compliance with ethical standards. All applicable institutional and international guidelines for research integrity and data confidentiality were followed.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors confirm that no AI tools were used to generate or assist in writing the manuscript. All content was produced and verified by the authors themselves.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

References

[1] Zhongao Zhai, Cong Chen, Yi Zhang, Temperature and deformation in laser and hybrid welding: a comprehensive review, ISSN 1526-6125, J. Manuf. Process. 152 (2025) 588–615, https://doi.org/10.1016/j.jmapro.2025.08.039.

[2] Akshansh Mishra, Abdelkader Miloudi, Eyob Messele Sefene, Chaaben Arroussi, Ismail Chekalil, Bassam Gamal Nasser Muthanna, Hybrid deep learning model to predict the ultimate tensile strength of friction stir welded joints, ISSN 0952-1976, Eng. Appl. Artif. Intell. 154 (2025) 111001, https://doi.org/10.1016/j.engappai.2025.111001.

[3] H. Danielewski, A. Skrzypczyk, Steel sheets laser lap joint welding—process analysis, Materials 13 (2020) 2258, https://doi.org/10.3390/ma13102258.

[4] A. Zwoliński, S. Samborski, J. Rzeczkowski, Laser technologies of welding, surfacing and regeneration of metals with HCP structure (Mg, Ti, Zr): state of the art, challenges and prospects, Materials 18 (22) (2025) 5237, https://doi.org/10.3390/ma18225237. PMID: 41304081; PMCID: PMC12654634.

[5] Q.-Y. Jin, J. Jung, J. Cheon, C. Ji, W. Lee, Melt pool simulation of dual laser beam-arc hybrid welding of aluminum alloy using finite element method, Materials 18 (1) (2025) 135, https://doi.org/10.3390/ma18010135.

[6] Dame Alemayehu Efa, Computational modeling and virtual analysis using a moving heat source to join AZ61A and AA7075 alloys with the application of a titanium alloy interlayer, ISSN 1350-4495, Infrared Phys. & Technol. 141 (2024) 105501, https://doi.org/10.1016/j.infrared.2024.105501.

[7] Xiangmeng Meng, Marcel Bachmann, Fan Yang, Michael Rethmeier, Toward prediction and insight of porosity formation in laser welding: a physics-informed deep learning framework, ISSN 1359-6454, Acta Mater. 286 (2025) 120740, https://doi.org/10.1016/j.actamat.2025.120740.

[8] Zhe Zhang, Qifan Zheng, Siyang Liu, Jian Guo, Analysis of Over-Melt Zone in Welding Temperature Field Based on CNN. Proceedings of the 2025 International Conference on Artificial Intelligence, Virtual Reality and Interaction Design (AIVRID '25), Association for Computing Machinery, New York, NY, USA, 2026, pp. 185–190, https://doi.org/10.1145/3777730.3777761.

[9] O. Falodun, S. Oke, M. Bodunrin, A comprehensive review of residual stresses in carbon steel welding: formation mechanisms, mitigation strategies, and advanced post-weld heat treatment techniques, Int J. Adv. Manuf. Technol. 136 (2025) 4107–4140, https://doi.org/10.1007/s00170-025-15088-8.

[10] L. Xie, W. Shi, T. Wu, M. Gong, D. Cai, S. Han, K. He, Effect of dynamic preheating on the thermal behavior and mechanical properties of laser-welded joints, Materials 15 (2022) 6159, https://doi.org/10.3390/ma15176159.

[11] T. Wu, W. Shi, L. Xie, M. Gong, J. Huang, Y. Xie, K. He, Effect of preheating temperature on geometry and mechanical properties of laser cladding-based stellite 6/WC coating, Materials 15 (2022) 3952.

[12] H. Liu, X. Du, H. Guo, J. Liu, P. Chen, H. Yang, J. Hao, Finite element analysis of effects of dynamic preheating on thermal behavior of multi-track and multi-layer laser cladding, Optik 228 (2021) 166194.

[13] J. Gao, C. Wu, Y. Hao, X. Xu, L. Guo, Numerical simulation and experimental investigation on three-dimensional modelling of single-track geometry and temperature evolution by laser cladding, Opt. Laser Technol. 129 (2020) 106287.

[14] X. Zou, Z. Yan, K. Zou, W. Liu, L. Song, S. Li, L. Cha, Grain refinement by dynamic recrystallization during laser direct energy deposition of 316L stainless steel under thermal cycles, J. Manuf. Process. 76 (2022) 646–655.

[15] C. Li, D. Zhang, X. Gao, H. Gao, X. Han, Numerical simulation method of the multi-field coupling mechanism for laser cladding 316L powder, Weld. World 66 (2022) 423–440.

[16] Yuelin Zhang, Xuchen Liu, Yi Zhou, Yanlong Shi, Influence of welding method on residual stress and metallography of a mild steel welded butt-joint plate, ISSN 0143-974X, J. Constr. Steel Res. 199 (2022) 107640, https://doi.org/10.1016/j.jcsr.2022.107640.

[17] Kangda Hao, Zhiyuan Gao, Junwei Huang, Lianyong Xu, Youqing Liu, Yongdian Han, Lei Zhao, Wenjin Ren, Comparisons of laser and laser-arc hybrid welded carbon steel with beam oscillation, ISSN 0030-3992, Opt. & Laser Technol. 157 (2023) 108787, https://doi.org/10.1016/j.optlastec.2022.108787.

[18] Xinglin Qu, Xiaofeng Li, Li Zhang, Denghao Yi, Jianhong Wang, Chengyang Wen, Zhanyong Zhao, Xingyu Gu, Yuan Lin, Bin Liu, Yuankui Cao, Peikang Bai, Effect of construction angles on the microstructure and mechanical properties of LPBF-fabricated 15-5 PH stainless steel, ISSN 0921-5093, Mater. Sci. Eng. A 900 (2024) 146423, https://doi.org/10.1016/j.msea.2024.146423.

[19] M. Rethmeier, A. Gumenyuk, M. Bachmann, High-power laser beam welding for thick section steels – new perspectives using electromagnetic systems, Sci. Technol. Weld. Join. 27 (1) (2022) 43–51, https://doi.org/10.1080/13621718.2021.1999763.

[20] Florian Fetzer, Christian Hagenlocher, Rudolf Weber, Thomas Graf, Geometry and stability of the capillary during deep-penetration laser welding of AlMgSi at high feed rates, ISSN 0030-3992, Opt. Laser Technol. 133 (2021) 106562, https://doi.org/10.1016/j.optlastec.2020.106562.

[21] M. Atwya, G. Panoutsos, In-situ porosity prediction in metal powder bed fusion additive manufacturing using spectral emissions: a prior-guided machine learning approach, J. Intell. Manuf. 35 (6) (2024) 2719–2742.

[22] Huan Liu, Wenxiu Fan, Ziyue Xu, Chao Sun, Yuna Wu, Jia Ju, Jiang Ma, Research progress of solid-phase welding technology between magnesium and dissimilar

metals, ISSN 2238-7854, J. Mater. Res. Technol. 38 (2025) 5409–5440, https://doi.org/10.1016/j.jmrt.2025.09.029.
[23] U.A. Khaliq, M.R. Muhamad, F. Yusof, S. Ibrahim, M.S.M. Isa, Z. Chen, G. Çam, A review on friction stir butt welding of aluminum with magnesium: a new insight on joining mechanisms by interfacial enhancement, J. Mater. Res. Technol. 27 (2023) 4595–4624.
[24] An. Xin Zhang, Feng Li, Peng Da Huo, Wen Tao Niu, Rong He Gao, Response mechanism of matrix microstructure evolution and mechanical behavior to Mg/Al composite plate by hard-plate accumulative roll bonding, ISSN 2238-7854, J. Mater. Res. Technol. 23 (2023) 3312–3321, https://doi.org/10.1016/j.jmrt.2023.01.227.